



Statistics and Data Science

StataNow 19.5
BE-Basic Edition

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Notes:

1. Unicode is supported; see [help unicode advice](#).
2. New update available; type `-update all-`

```
1 . doedit "C:\Users\yoaga\OneDrive\Escritorio\Survival\PruebaFinal_2\CodigoFinal_Rev2.do"
2 . *****
3 .
4 . ** Death Analysis
5 .
6 . ** Based on T. Bishop' approach and modified by Johana Ramos
7 .
8 . *****
9 .
10 .
11 .
12 .
13 .
14 . clear
15 .
16 . cd "C:\Users\yoaga\OneDrive\Escritorio\Survival\PruebaFinal_2"
    C:\Users\yoaga\OneDrive\Escritorio\Survival\PruebaFinal_2
17 .
18 . use amdata.dta
19 .
20 . log using overall_survival.log, replace
```

name: <unnamed>
log: C:\Users\yoaga\OneDrive\Escritorio\Survival\PruebaFinal_2\overall_survival.log
log type: text
opened on: 18 Sep 2025, 22:24:03

```

21 .
22 .
23 .
24 . generate t = dthdate
    (70 missing values generated)

25 .
26 . replace t = lndate if dthdate == .
    (70 real changes made)

27 .
28 .
29 .
30 . egen tag = tag(tumor_sample_barcode)

31 .
32 . tab tag

```

tag(tumor_s ample_barco de)	Freq.	Percent	Cum.
1	85	100.00	100.00
Total	85	100.00	

```

33 .
34 .
35 .
36 . drop tag

37 .
38 . generate Mutation = upper(mutation)

39 .
40 . replace Mutation = " WT" if mutation == "qwt"
    (43 real changes made)

41 .
42 .
43 .
44 . tab Mutation death, row all

```

Key
<i>frequency</i> <i>row percentage</i>

Mutation	death		Total
	0	1	
WT	39 90.70	4 9.30	43 100.00
BRAF	7 63.64	4 36.36	11 100.00
KIT	10 90.91	1 9.09	11 100.00
MULTIHIT	0 0.00	1 100.00	1 100.00
NF1	5 71.43	2 28.57	7 100.00

NRAS	9 75.00	3 25.00	12 100.00
Total	70 82.35	15 17.65	85 100.00

Pearson chi2(5) = 10.9539 Pr = 0.052
 Likelihood-ratio chi2(5) = 9.6104 Pr = 0.087
 Cramér's V = 0.3590
 gamma = 0.3504 ASE = 0.168
 Kendall's tau-b = 0.1782 ASE = 0.096

```

45 .
46 .
47 .
48 . tab imut death, all exact row

```

Key
<i>frequency</i>
<i>row percentage</i>

imut	death		Total
	0	1	
0	39 90.70	4 9.30	43 100.00
1	31 73.81	11 26.19	42 100.00
Total	70 82.35	15 17.65	85 100.00

Pearson chi2(1) = 4.1698 Pr = 0.041
 Likelihood-ratio chi2(1) = 4.3015 Pr = 0.038
 Cramér's V = 0.2215
 gamma = 0.5515 ASE = 0.220
 Kendall's tau-b = 0.2215 ASE = 0.100
 Fisher's exact = 0.050
 1-sided Fisher's exact = 0.038

```

49 .
50 .
51 .
52 . generate cstage = stage

53 .
54 . destring cstage, replace ignore("NA")
    cstage already numeric; no replace

55 .

```

```

56 . replace cstage = . if cstage == 0
    (1 real change made, 1 to missing)

57 .
58 .
59 .
60 . replace cstage = min(cstage,3)
    (6 real changes made)

61 .
62 . replace cstage = max(cstage,2)
    (10 real changes made)

63 .
64 .
65 .
66 . generate ncluster = .
    (85 missing values generated)

67 .
68 . replace ncluster = 1 if cluster == "2"
    (16 real changes made)

69 .
70 . replace ncluster = 2 if cluster == "1"
    (14 real changes made)

71 .
72 . replace ncluster = 3 if cluster == "3"
    (14 real changes made)

73 .
74 .
75 .
76 . tab ncluster death, row all

```

Key
<i>frequency</i>
<i>row percentage</i>

ncluster	death		Total
	0	1	
1	9 56.25	7 43.75	16 100.00
2	14 100.00	0 0.00	14 100.00
3	11 78.57	3 21.43	14 100.00
Total	34 77.27	10 22.73	44 100.00

```

Pearson chi2(2) = 8.1576 Pr = 0.017
Likelihood-ratio chi2(2) = 10.6862 Pr = 0.005
Cramér's V = 0.4306
gamma = -0.4344 ASE = 0.290
Kendall's tau-b = -0.2265 ASE = 0.158

```

```

77 .
78 .
79 .
80 .
81 .
82 . generate follow = t - dxdate

83 .
84 .
85 .
86 . tab Mutation death, all exact row

```

Key
<i>frequency</i>
<i>row percentage</i>

Enumerating sample-space combinations:

```

stage 6: enumerations = 1
stage 5: enumerations = 2
stage 4: enumerations = 8
stage 3: enumerations = 40
stage 2: enumerations = 102
stage 1: enumerations = 0

```

Mutation	death		Total
	0	1	
WT	39 90.70	4 9.30	43 100.00
BRAF	7 63.64	4 36.36	11 100.00
KIT	10 90.91	1 9.09	11 100.00
MULTIHIT	0 0.00	1 100.00	1 100.00
NF1	5 71.43	2 28.57	7 100.00
NRAS	9 75.00	3 25.00	12 100.00
Total	70 82.35	15 17.65	85 100.00

```

Pearson chi2(5) = 10.9539 Pr = 0.052
Likelihood-ratio chi2(5) = 9.6104 Pr = 0.087
Cramér's V = 0.3590
gamma = 0.3504 ASE = 0.168
Kendall's tau-b = 0.1782 ASE = 0.096
Fisher's exact = 0.042

```

87 .
88 .
89 .

90 . ltable follow death, test notable graph overlay xlab(0(400)4000) ylab(0(0.2)1) by(Mutation) xtitle("Since recruitment

Likelihood-ratio test statistic of homogeneity (group=Mutation):
chi2(5) = **8.9510193**, P = **.11103386**

Logrank test of homogeneity (group=Mutation):

Equality of survivor functions
Log-rank test

Mutation	Observed events	Expected events
WT	4	7.43
BRAF	4	2.02
KIT	1	1.85
MULTIHIT	1	0.05
NF1	2	1.17
NRAS	3	2.48
Total	15	15.00

chi2(5) = **21.88**
Pr>chi2 = **0.0006**

91 .
92 . graph export ltable_deathmutation.pdf, as(pdf) replace
file **ltable_deathmutation.pdf** saved as PDF format

93 .
94 .
95 .

96 . ltable follow death, test notable graph overlay xlab(0(400)4000) ylab(0(0.2)1) by(imut) xtitle("Since recruitment (day

Likelihood-ratio test statistic of homogeneity (group=imut):
chi2(1) = **3.4855042**, P = **.0619085**

Logrank test of homogeneity (group=imut):

Equality of survivor functions
Log-rank test

imut	Observed events	Expected events
0	4	7.43
1	11	7.57
Total	15	15.00

chi2(1) = **3.13**
Pr>chi2 = **0.0766**

```

97 .
98 . graph export ltable_deathmutation.pdf, as(pdf) replace
    file ltable_deathmutation.pdf saved as PDF format

99 .
100 .
101 .
102 . ltable follow death, test notable graph overlay xlab(0(400)4000) ylab(0(0.2)1) by(ncluster) xtitle("Since recruitment

```

Likelihood-ratio test statistic of homogeneity (group=ncluster):
 $\chi^2(2) = 13.783539$, $P = .00101611$

Logrank test of homogeneity (group=ncluster):

Equality of survivor functions
 Log-rank test

ncluster	Observed events	Expected events
1	7	3.10
2	0	3.99
3	3	2.91
Total	10	10.00

$\chi^2(2) = 8.99$
 $Pr > \chi^2 = 0.0111$

```

103 .
104 . graph export ltable_deathmutation.pdf, as(pdf) replace
    file ltable_deathmutation.pdf saved as PDF format

105 .
106 .
107 .
108 . stset t, failure(death) origin(time dxdate) enter(time condate)

```

Survival-time data settings

Failure event: **death!=0 & death<.**
 Observed time interval: **(origin, t]**
 Enter on or after: **time condate**
 Exit on or before: **failure**
 Time for analysis: **(time-origin)**
 Origin: **time dxdate**

85 total observations
 0 exclusions

85 observations remaining, representing
 15 failures in single-record/single-failure data
 97,084 total analysis time at risk and under observation

At risk from t =	0
Earliest observed entry t =	0
Last observed exit t =	5,063

```

109 .
110 .
111 .
112 . stcox imut

```

```

      Failure _d: death
      Analysis time _t: (t-origin)
      Origin: time dxdate
      Enter on or after: time condate

```

```

Iteration 0: Log likelihood = -53.343919
Iteration 1: Log likelihood = -52.576297
Iteration 2: Log likelihood = -52.571055
Iteration 3: Log likelihood = -52.571054
Refining estimates:
Iteration 0: Log likelihood = -52.571054

```

Cox regression with no ties

```

No. of subjects =      85      Number of obs =      85
No. of failures =      15
Time at risk    = 97,084

Log likelihood = -52.571054      LR chi2(1)    =    1.55
                                Prob > chi2    = 0.2138

```

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
imut	2.003101	1.172107	1.19	0.235	.6362463	6.306383

```

113 .
114 .
115 .
116 . xi: logistic imut sex age i.cstage dxdate Q2AMR
      i.cstage      _Icstage_2-3      (naturally coded; _Icstage_2 omitted)

```

```

Logistic regression      Number of obs =    73
                        LR chi2(5)    =   13.16
                        Prob > chi2    = 0.0219
Log likelihood = -43.849075      Pseudo R2    = 0.1305

```

imut	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
sex	.2619569	.1417737	-2.48	0.013	.0906883	.7566736
age	.9646099	.0228448	-1.52	0.128	.9208582	1.01044
_Icstage_3	1.597506	.8650562	0.87	0.387	.5527302	4.617126
dxdate	1.000281	.0003449	0.82	0.415	.9996054	1.000957
Q2AMR	.0625818	.0849547	-2.04	0.041	.0043746	.8952685
_cons	.2706365	1.913251	-0.18	0.853	2.60e-07	281781.8

Note: **_cons** estimates baseline odds.

```

117 .

```



```

118 .
119 .
120 . xi: logistic i.Mutation sex age i.cstage dxdate Q2AMR
      i.Mutation      _IMutation_1-6      (_IMutation_1 for Mutation==WT omitted)
      i.cstage        _Icstage_2-3        (naturally coded; _Icstage_2 omitted)
      note: _IMutation_3 != 0 predicts failure perfectly;
            _IMutation_3 omitted and 9 obs not used.

      note: _IMutation_4 != 0 predicts failure perfectly;
            _IMutation_4 omitted and 1 obs not used.

      note: _IMutation_5 != 0 predicts failure perfectly;
            _IMutation_5 omitted and 5 obs not used.

      note: _IMutation_6 != 0 predicts failure perfectly;
            _IMutation_6 omitted and 10 obs not used.

```

```

Logistic regression                                Number of obs =      48
                                                    LR chi2(5)      =   11.87
                                                    Prob > chi2     =  0.0367
Log likelihood = -17.230535                      Pseudo R2      =  0.2561

```

_IMutation_2	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
_IMutation_3	1 (omitted)					
_IMutation_4	1 (omitted)					
_IMutation_5	1 (omitted)					
_IMutation_6	1 (omitted)					
sex	.132809	.142368	-1.88	0.060	.0162465	1.085662
age	1.020621	.0471542	0.44	0.659	.9322618	1.117356
_Icstage_3	2.284501	2.058127	0.92	0.359	.3907836	13.35507
dxdate	1.000645	.0008105	0.80	0.426	.9990572	1.002234
Q2AMR	.0060979	.0139796	-2.22	0.026	.0000682	.5452434
_cons	5.42e-06	.0000898	-0.73	0.464	4.22e-20	6.97e+08

Note: _cons estimates baseline odds.

```

121 .
122 .
123 .
124 . foreach gn in "brf" "hras" "kit" "kras" "nf1" "nras" "mut" {
      2.
125 . display "`gn'"
      3.
126 . tab1 i`gn'
      4.
127 . xi: stcox i`gn' sex age i.cstage
      5.
128 . xi: stcox i`gn' sex age i.cstage Q2AMR
      6.
129 . xi: logistic recurrence imut sex age i.cstage dxdate Q2AMR
      7.
130 . }
      brf

```

-> tabulation of ibraf

ibraf	Freq.	Percent	Cum.
0	43	79.63	79.63
1	11	20.37	100.00

```

Total      54      100.00
i.cstage   _Icstage_2-3      (naturally coded; _Icstage_2 omitted)

```

Failure _d: death
 Analysis time _t: (t-origin)
 Origin: time dxdate
 Enter on or after: time condate

Iteration 0: Log likelihood = -24.589624
 Iteration 1: Log likelihood = -21.271217
 Iteration 2: Log likelihood = -21.116569
 Iteration 3: Log likelihood = -21.116415
 Refining estimates:
 Iteration 0: Log likelihood = -21.116415

Cox regression with no ties

No. of subjects = 54 Number of obs = 54
 No. of failures = 8
 Time at risk = 59,092
 Log likelihood = -21.116415 LR chi2(4) = 6.95
 Prob > chi2 = 0.1387

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
ibraf	3.949793	4.113829	1.32	0.187	.5128877	30.4177
sex	3.580555	3.807455	1.20	0.230	.4454594	28.78012
age	1.065517	.0416156	1.62	0.104	.9869956	1.150285
_Icstage_3	1.517638	1.327371	0.48	0.633	.2733268	8.426637

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure _d: death
 Analysis time _t: (t-origin)
 Origin: time dxdate
 Enter on or after: time condate

Iteration 0: Log likelihood = -17.222925
 Iteration 1: Log likelihood = -13.683475
 Iteration 2: Log likelihood = -13.55685
 Iteration 3: Log likelihood = -13.556367
 Iteration 4: Log likelihood = -13.556367
 Refining estimates:
 Iteration 0: Log likelihood = -13.556367

Cox regression with no ties

No. of subjects = 48 Number of obs = 48
 No. of failures = 6
 Time at risk = 51,212
 Log likelihood = -13.556367 LR chi2(5) = 7.33
 Prob > chi2 = 0.1970

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
ibraf	4.716056	6.309751	1.16	0.246	.3425557	64.92721
sex	1.997863	2.692951	0.51	0.608	.1423041	28.04877
age	1.100602	.0642854	1.64	0.101	.9815492	1.234094
_Icstage_3	2.573783	2.844546	0.86	0.392	.295001	22.45538
Q2AMR	.1093885	.2657026	-0.91	0.362	.0009363	12.7798

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Logistic regression Number of obs = 73
 LR chi2(6) = 18.11
 Prob > chi2 = 0.0060
 Log likelihood = -41.484649 Pseudo R2 = 0.1791

recurrence	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
imut	5.310556	3.325634	2.67	0.008	1.556293	18.12127
sex	.9844121	.5578017	-0.03	0.978	.3242328	2.988801
age	1.016565	.0244662	0.68	0.495	.9697256	1.065667
_Icstage_3	3.545433	1.973259	2.27	0.023	1.191027	10.554
dxdate	1.000076	.000339	0.23	0.822	.9994121	1.000741
Q2AMR	21.06773	33.44283	1.92	0.055	.9384777	472.9461
_cons	.0018219	.0129485	-0.89	0.375	1.63e-09	2042.56

Note: **_cons** estimates baseline odds.
hras

-> tabulation of ihras

ihras	Freq.	Percent	Cum.
0	43	100.00	100.00
Total	43	100.00	

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure **_d**: death
Analysis time **_t**: (t-origin)
Origin: time dxdate
Enter on or after: time conddate

note: **ihras** omitted because of collinearity.

Iteration 0: Log likelihood = -10.485759

Iteration 1: Log likelihood = -9.0248097

Iteration 2: Log likelihood = -8.7964222

Iteration 3: Log likelihood = -8.7864807

Iteration 4: Log likelihood = -8.7864593

Refining estimates:

Iteration 0: Log likelihood = -8.7864593

Cox regression with no ties

No. of subjects = 43

Number of obs = 43

No. of failures = 4

Time at risk = 43,171

LR chi2(3) = 3.40

Log likelihood = -8.7864593

Prob > chi2 = 0.3342

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
ihras	1 (omitted)					
sex	2.087671	2.520788	0.61	0.542	.1958242	22.25654
age	1.108697	.0851452	1.34	0.179	.9537681	1.288793
_Icstage_3	3.092662	4.271799	0.82	0.414	.2063451	46.35225

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure **_d**: death
Analysis time **_t**: (t-origin)
Origin: time dxdate
Enter on or after: time conddate

note: **ihras** omitted because of collinearity.

Iteration 0: Log likelihood = **-7.1163941**

Iteration 1: Log likelihood = **-6.1739134**

Iteration 2: Log likelihood = **-6.0245447**

Iteration 3: Log likelihood = **-6.020045**

Iteration 4: Log likelihood = **-6.0200409**

Refining estimates:

Iteration 0: Log likelihood = **-6.0200409**

Cox regression with no ties

No. of subjects = **39**

Number of obs = **39**

No. of failures = **3**

Time at risk = **38,459**

LR chi2(4) = **2.19**

Log likelihood = **-6.0200409**

Prob > chi2 = **0.7004**

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
ihras	1 (omitted)					
sex	1.511843	1.928657	0.32	0.746	.1240593	18.424
age	1.119358	.1052058	1.20	0.230	.9310364	1.345772
_Icstage_3	3.470709	5.996797	0.72	0.471	.1174049	102.6007
Q2AMR	.3499607	1.030506	-0.36	0.721	.0010903	112.33

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Logistic regression

Number of obs = **73**

LR chi2(6) = **18.11**

Prob > chi2 = **0.0060**

Log likelihood = **-41.484649**

Pseudo R2 = **0.1791**

recurrence	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
imut	5.310556	3.325634	2.67	0.008	1.556293	18.12127
sex	.9844121	.5578017	-0.03	0.978	.3242328	2.988801
age	1.016565	.0244662	0.68	0.495	.9697256	1.065667
_Icstage_3	3.545433	1.973259	2.27	0.023	1.191027	10.554
dxdate	1.000076	.000339	0.23	0.822	.9994121	1.000741
Q2AMR	21.06773	33.44283	1.92	0.055	.9384777	472.9461
_cons	.0018219	.0129485	-0.89	0.375	1.63e-09	2042.56

Note: **_cons** estimates baseline odds.
kit

-> tabulation of ikit

ikit	Freq.	Percent	Cum.
0	43	79.63	79.63
1	11	20.37	100.00
Total	54	100.00	

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure _d: **death**

Analysis time _t: **(t-origin)**

Origin: **time dxdate**

Enter on or after: **time condade**

Logistic regression	Number of obs =	73
	LR chi2(6) =	18.11
	Prob > chi2 =	0.0060
Log likelihood = -41.484649	Pseudo R2 =	0.1791

recurrence	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
imut	5.310556	3.325634	2.67	0.008	1.556293	18.12127
sex	.9844121	.5578017	-0.03	0.978	.3242328	2.988801
age	1.016565	.0244662	0.68	0.495	.9697256	1.065667
_Icstage_3	3.545433	1.973259	2.27	0.023	1.191027	10.554
dxdate	1.000076	.000339	0.23	0.822	.9994121	1.000741
Q2AMR	21.06773	33.44283	1.92	0.055	.9384777	472.9461
_cons	.0018219	.0129485	-0.89	0.375	1.63e-09	2042.56

Note: **_cons** estimates baseline odds.
kras

-> tabulation of ikras

ikras	Freq.	Percent	Cum.
0	43	100.00	100.00
Total	43	100.00	

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure **_d**: death
Analysis time **_t**: (t-origin)
Origin: time dxdate
Enter on or after: time conddate

note: **ikras** omitted because of collinearity.

Iteration 0: Log likelihood = -10.485759
Iteration 1: Log likelihood = -9.0248097
Iteration 2: Log likelihood = -8.7964222
Iteration 3: Log likelihood = -8.7864807
Iteration 4: Log likelihood = -8.7864593
Refining estimates:
Iteration 0: Log likelihood = -8.7864593

Cox regression with no ties

No. of subjects =	43	Number of obs =	43
No. of failures =	4		
Time at risk =	43,171		
Log likelihood =	-8.7864593	LR chi2(3) =	3.40
		Prob > chi2 =	0.3342

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
ikras	1 (omitted)					
sex	2.087671	2.520788	0.61	0.542	.1958242	22.25654
age	1.108697	.0851452	1.34	0.179	.9537681	1.288793
_Icstage_3	3.092662	4.271799	0.82	0.414	.2063451	46.35225

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure **_d**: death
Analysis time **_t**: (t-origin)
Origin: time dxdate
Enter on or after: time conddate

note: **ikras** omitted because of collinearity.

Iteration 0: Log likelihood = **-7.1163941**

Iteration 1: Log likelihood = **-6.1739134**

Iteration 2: Log likelihood = **-6.0245447**

Iteration 3: Log likelihood = **-6.020045**

Iteration 4: Log likelihood = **-6.0200409**

Refining estimates:

Iteration 0: Log likelihood = **-6.0200409**

Cox regression with no ties

No. of subjects = **39**

Number of obs = **39**

No. of failures = **3**

Time at risk = **38,459**

LR chi2(4) = **2.19**

Log likelihood = **-6.0200409**

Prob > chi2 = **0.7004**

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
ikras	1 (omitted)					
sex	1.511843	1.928657	0.32	0.746	.1240593	18.424
age	1.119358	.1052058	1.20	0.230	.9310364	1.345772
_Icstage_3	3.470709	5.996797	0.72	0.471	.1174049	102.6007
Q2AMR	.3499607	1.030506	-0.36	0.721	.0010903	112.33

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Logistic regression

Number of obs = **73**

LR chi2(6) = **18.11**

Prob > chi2 = **0.0060**

Log likelihood = **-41.484649**

Pseudo R2 = **0.1791**

recurrence	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
imut	5.310556	3.325634	2.67	0.008	1.556293	18.12127
sex	.9844121	.5578017	-0.03	0.978	.3242328	2.988801
age	1.016565	.0244662	0.68	0.495	.9697256	1.065667
_Icstage_3	3.545433	1.973259	2.27	0.023	1.191027	10.554
dxdate	1.000076	.000339	0.23	0.822	.9994121	1.000741
Q2AMR	21.06773	33.44283	1.92	0.055	.9384777	472.9461
_cons	.0018219	.0129485	-0.89	0.375	1.63e-09	2042.56

Note: **_cons** estimates baseline odds.

nf1

-> tabulation of inf1

inf1	Freq.	Percent	Cum.
0	43	86.00	86.00
1	7	14.00	100.00
Total	50	100.00	

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure _d: **death**

Analysis time _t: **(t-origin)**

Origin: **time dxdate**

Enter on or after: **time conddate**

Iteration 0: Log likelihood = -16.916305
 Iteration 1: Log likelihood = -14.401477
 Iteration 2: Log likelihood = -14.158033
 Iteration 3: Log likelihood = -14.155948
 Iteration 4: Log likelihood = -14.155947
 Refining estimates:
 Iteration 0: Log likelihood = -14.155947

Cox regression with no ties

No. of subjects = 50 Number of obs = 50
 No. of failures = 6
 Time at risk = 48,731
 LR chi2(4) = 5.52
 Prob > chi2 = 0.2379
 Log likelihood = -14.155947

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
inf1	3.5397	3.447827	1.30	0.194	.5246356	23.88224
sex	3.888354	4.328121	1.22	0.222	.4388284	34.45379
age	1.038928	.0506883	0.78	0.434	.944183	1.14318
_Icstage_3	3.183061	3.524775	1.05	0.296	.3632985	27.88857

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure _d: death
 Analysis time _t: (t-origin)
 Origin: time dxdate
 Enter on or after: time condate

Iteration 0: Log likelihood = -13.038795
 Iteration 1: Log likelihood = -10.454004
 Iteration 2: Log likelihood = -9.9022941
 Iteration 3: Log likelihood = -9.8896936
 Iteration 4: Log likelihood = -9.889682
 Refining estimates:
 Iteration 0: Log likelihood = -9.889682

Cox regression with no ties

No. of subjects = 44 Number of obs = 44
 No. of failures = 5
 Time at risk = 43,294
 LR chi2(5) = 6.30
 Prob > chi2 = 0.2783
 Log likelihood = -9.889682

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
inf1	5.470726	7.498192	1.24	0.215	.3727226	80.29789
sex	3.037657	3.507916	0.96	0.336	.3159095	29.20887
age	1.0846	.0883574	1.00	0.319	.9245409	1.272369
_Icstage_3	5.820538	7.935493	1.29	0.196	.4022244	84.22827
Q2AMR	.1303938	.3558782	-0.75	0.455	.0006196	27.44133

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Logistic regression Number of obs = 73
 LR chi2(6) = 18.11
 Prob > chi2 = 0.0060
 Pseudo R2 = 0.1791
 Log likelihood = -41.484649

recurrence	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
imut	5.310556	3.325634	2.67	0.008	1.556293	18.12127
sex	.9844121	.5578017	-0.03	0.978	.3242328	2.988801
age	1.016565	.0244662	0.68	0.495	.9697256	1.065667
_Icstage_3	3.545433	1.973259	2.27	0.023	1.191027	10.554
dxdate	1.000076	.000339	0.23	0.822	.9994121	1.000741
Q2AMR	21.06773	33.44283	1.92	0.055	.9384777	472.9461
_cons	.0018219	.0129485	-0.89	0.375	1.63e-09	2042.56

Note: **_cons** estimates baseline odds.
nras

-> tabulation of inras

inras	Freq.	Percent	Cum.
0	43	78.18	78.18
1	12	21.82	100.00
Total	55	100.00	

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure _d: death
Analysis time _t: (t-origin)
Origin: time dxdate
Enter on or after: time condate

Iteration 0: Log likelihood = -20.71374
Iteration 1: Log likelihood = -19.652707
Iteration 2: Log likelihood = -19.620181
Iteration 3: Log likelihood = -19.620142
Refining estimates:
Iteration 0: Log likelihood = -19.620142

Cox regression with no ties

No. of subjects = 55 Number of obs = 55
No. of failures = 7
Time at risk = 61,283
Log likelihood = -19.620142 LR chi2(4) = 2.19
Prob > chi2 = 0.7014

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
inras	3.702205	4.004267	1.21	0.226	.4444366	30.83978
sex	1.447616	1.314946	0.41	0.684	.2440408	8.587058
age	1.030597	.0439152	0.71	0.479	.9480209	1.120365
_Icstage_3	1.955781	1.662939	0.79	0.430	.3694625	10.35309

i.cstage _Icstage_2-3 (naturally coded; _Icstage_2 omitted)

Failure _d: death
Analysis time _t: (t-origin)
Origin: time dxdate
Enter on or after: time condate

Iteration 0: Log likelihood = -16.549229
Iteration 1: Log likelihood = -13.929776
Iteration 2: Log likelihood = -13.264579
Iteration 3: Log likelihood = -13.232312
Iteration 4: Log likelihood = -13.23221
Refining estimates:
Iteration 0: Log likelihood = -13.23221

Cox regression with no ties

No. of subjects =	49	Number of obs =	49
No. of failures =	6		
Time at risk =	52,374		
		LR chi2(5) =	6.63
Log likelihood =	-13.23221	Prob > chi2 =	0.2493

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
inras	103.1896	238.4296	2.01	0.045	1.113913	9559.17
sex	.458614	.5113566	-0.70	0.484	.0515649	4.078877
age	1.156162	.094888	1.77	0.077	.9843719	1.357932
_Icstage_3	9.103087	12.5225	1.61	0.108	.6141129	134.9364
Q2AMR	.0638281	.1605489	-1.09	0.274	.0004613	8.832121

i.cstage	_Icstage_2-3	(naturally coded; _Icstage_2 omitted)
----------	--------------	---------------------------------------

Logistic regression	Number of obs =	73
	LR chi2(6) =	18.11
	Prob > chi2 =	0.0060
Log likelihood = -41.484649	Pseudo R2 =	0.1791

recurrence	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
imut	5.310556	3.325634	2.67	0.008	1.556293	18.12127
sex	.9844121	.5578017	-0.03	0.978	.3242328	2.988801
age	1.016565	.0244662	0.68	0.495	.9697256	1.065667
_Icstage_3	3.545433	1.973259	2.27	0.023	1.191027	10.554
dxdate	1.000076	.000339	0.23	0.822	.9994121	1.000741
Q2AMR	21.06773	33.44283	1.92	0.055	.9384777	472.9461
_cons	.0018219	.0129485	-0.89	0.375	1.63e-09	2042.56

Note: **_cons** estimates baseline odds.
mut

-> tabulation of imut

imut	Freq.	Percent	Cum.
0	43	50.59	50.59
1	42	49.41	100.00
Total	85	100.00	

i.cstage	_Icstage_2-3	(naturally coded; _Icstage_2 omitted)
----------	--------------	---------------------------------------

```

Failure _d: death
Analysis time _t: (t-origin)
Origin: time dxdate
Enter on or after: time condate

```

```
Iteration 0: Log likelihood = -53.343919
Iteration 1: Log likelihood = -50.061299
Iteration 2: Log likelihood = -49.993146
Iteration 3: Log likelihood = -49.993084
Refining estimates:
Iteration 0: Log likelihood = -49.993084
```

Cox regression with no ties


```

131 .
132 . destring cluster, replace ignore("NA")
    cluster: characters N A removed; replaced as byte
    (41 missing values generated)

```

```

133 .
134 . xi: stcox i.cluster
    i.cluster      _Icluster_1-3      (naturally coded; _Icluster_1 omitted)

```

```

    Failure _d: death
    Analysis time _t: (t-origin)
    Origin: time dxdate
    Enter on or after: time conddate

```

```

Iteration 0: Log likelihood = -29.102558
Iteration 1: Log likelihood = -26.128086
Iteration 2: Log likelihood = -25.577554
Iteration 3: Log likelihood = -25.407849
Iteration 4: Log likelihood = -25.3476
Iteration 5: Log likelihood = -25.325678
Iteration 6: Log likelihood = -25.317645
Iteration 7: Log likelihood = -25.314694
Iteration 8: Log likelihood = -25.313609
Iteration 9: Log likelihood = -25.31321
Iteration 10: Log likelihood = -25.313064
Iteration 11: Log likelihood = -25.31301
Iteration 12: Log likelihood = -25.31299
Iteration 13: Log likelihood = -25.312982
Iteration 14: Log likelihood = -25.31298
Iteration 15: Log likelihood = -25.312979
Iteration 16: Log likelihood = -25.312978
Iteration 17: Log likelihood = -25.312978
Iteration 18: Log likelihood = -25.312978
Iteration 19: Log likelihood = -25.312978
Iteration 20: Log likelihood = -25.312978
Iteration 21: Log likelihood = -25.312978
Refining estimates:
Iteration 0: Log likelihood = -25.312978

```

Cox regression with no ties

```

No. of subjects =      44      Number of obs =      44
No. of failures =      10
Time at risk    = 50,705

Log likelihood = -25.312978      LR chi2(1)    =      7.58
                                Prob > chi2    = 0.0059

```

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
_Icluster_2	3.63e+09	2.52e+09	31.70	0.000	9.31e+08	1.42e+10
_Icluster_3	1.71e+09	.	.	.	.	.

```

135 .

```

```

136 . xi: stcox i.cluster sex age i.cstage
      i.cluster      _Icluster_1-3      (naturally coded; _Icluster_1 omitted)
      i.cstage       _Icstage_2-3      (naturally coded; _Icstage_2 omitted)

```

```

      Failure _d: death
      Analysis time _t: (t-origin)
      Origin: time dxdate
      Enter on or after: time conddate

```

```

Iteration 0: Log likelihood = -29.102558
Iteration 1: Log likelihood = -25.53848
Iteration 2: Log likelihood = -24.969047
Iteration 3: Log likelihood = -24.797632
Iteration 4: Log likelihood = -24.7368
Iteration 5: Log likelihood = -24.714667
Iteration 6: Log likelihood = -24.706558
Iteration 7: Log likelihood = -24.703579
Iteration 8: Log likelihood = -24.702484
Iteration 9: Log likelihood = -24.702081
Iteration 10: Log likelihood = -24.701933
Iteration 11: Log likelihood = -24.701879
Iteration 12: Log likelihood = -24.701858
Iteration 13: Log likelihood = -24.701851
Iteration 14: Log likelihood = -24.701848
Iteration 15: Log likelihood = -24.701847
Iteration 16: Log likelihood = -24.701847
Iteration 17: Log likelihood = -24.701847
Iteration 18: Log likelihood = -24.701847
Iteration 19: Log likelihood = -24.701847
Iteration 20: Log likelihood = -24.701847
Iteration 21: Log likelihood = -24.701847
Refining estimates:
Iteration 0: Log likelihood = -24.701847

```

Cox regression with no ties

```

No. of subjects =      44      Number of obs =      44
No. of failures =      10
Time at risk    = 50,705

Log likelihood = -24.701847      LR chi2(4)    =      8.80
                                Prob > chi2    = 0.0663

```

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
_Icluster_2	3.56e+09	2.68e+09	29.23	0.000	8.14e+08	1.55e+10
_Icluster_3	1.79e+09	.	.	.	.	.
sex	1.486804	1.032671	0.57	0.568	.3811052	5.800461
age	1.007615	.0298154	0.26	0.798	.9508398	1.06778
_Icstage_3	1.766714	1.358867	0.74	0.459	.3912601	7.977506

```

137 .
138 . xi: stcox i.cluster sex age i.cstage Q2AMR
      i.cluster      _Icluster_1-3      (naturally coded; _Icluster_1 omitted)
      i.cstage       _Icstage_2-3      (naturally coded; _Icstage_2 omitted)

```

```

      Failure _d: death
      Analysis time _t: (t-origin)
      Origin: time dxdate
      Enter on or after: time conddate

```

```
Iteration 0: Log likelihood = -21.406136
Iteration 1: Log likelihood = -14.293651
Iteration 2: Log likelihood = -13.268917
Iteration 3: Log likelihood = -13.053423
Iteration 4: Log likelihood = -12.998267
Iteration 5: Log likelihood = -12.979635
Iteration 6: Log likelihood = -12.97284
Iteration 7: Log likelihood = -12.970348
Iteration 8: Log likelihood = -12.969432
Iteration 9: Log likelihood = -12.969095
Iteration 10: Log likelihood = -12.968971
Iteration 11: Log likelihood = -12.968925
Iteration 12: Log likelihood = -12.968909
Iteration 13: Log likelihood = -12.968902
Iteration 14: Log likelihood = -12.9689
Iteration 15: Log likelihood = -12.968899
Iteration 16: Log likelihood = -12.968899
Iteration 17: Log likelihood = -12.968899
Iteration 18: Log likelihood = -12.968899
Iteration 19: Log likelihood = -12.968899
Iteration 20: Log likelihood = -12.968899
Iteration 21: Log likelihood = -12.968899
Refining estimates:
Iteration 0: Log likelihood = -12.968899
```

Cox regression with no ties

No. of subjects =	37	Number of obs =	37
No. of failures =	8		
Time at risk =	42,130		
Log likelihood =	-12.968899	LR chi2(5) =	16.87
		Prob > chi2 =	0.0047

_t	Haz. ratio	Std. err.	z	P> z	[95% conf. interval]	
_Icluster_2	1.20e+09	1.16e+09	21.70	0.000	1.82e+08	7.95e+09
_Icluster_3	1.32e+09	.	.	.	.	.
sex	.9881788	1.030601	-0.01	0.991	.1279652	7.63096
age	.9821657	.0508847	-0.35	0.728	.88733	1.087137
_Icstage_3	1.702749	1.633946	0.55	0.579	.2596234	11.16754
Q2AMR	.0000112	.0000612	-2.08	0.037	2.44e-10	.5117382

```
139 .
140 . log close
      name: <unnamed>
      log: C:\Users\yoaga\OneDrive\Escritorio\Survival\PruebaFinal_2\overall_survival.log
      log type: text
      closed on: 18 Sep 2025, 22:24:16
```

141 .
142 .
143 .

144 .

145 .

146 . ltable follow death, test notable graph overlay xlab(0(400)4000) ylab(0(0.2)1) by(ncluster) xtitle("Since recruitment

Likelihood-ratio test statistic of homogeneity (group=ncluster):

chi2(2) = **13.783539**, P = **.00101611**

Logrank test of homogeneity (group=ncluster):

Equality of survivor functions

Log-rank test

ncluster	Observed events	Expected events
1	7	3.10
2	0	3.99
3	3	2.91
Total	10	10.00

chi2(2) = **8.99**Pr>chi2 = **0.0111**

147 .

148 . graph export ltable_deathRNAcluster.pdf, as(pdf) replace
file **ltable_deathRNAcluster.pdf** saved as PDF format

149 .